

Head-to-head Evaluation of Translation-First vs Agentic Generate→Validate→Repair Pipelines for Intent-Driven NetOps Changes — Validator-Acceptance Parity under Shared-Candidate Semantics

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Abstract—We report a fully preregistered, artifact-archived paired experiment (CAND-2026-001-prereg-v1) that compared two operational architectures for intent→configuration NetOps changes across heterogeneous vendor CLIs and Mininet-derived topologies: (A) translation-first (constrained vendor DSL → formal verifier → solver) and (B) agentic generate→validate→repair (hosted LLM + deterministic validators). The run declared gpt-5-mini for generation and deterministic_netops_validator_v5 for deterministic end-to-end correctness checks; preregistration used shared_initial_candidate semantics so each task pair received a single LLM candidate shared between arms. Execution completed 240 episodes (120 paired tasks) with model_calls_used = 120 (one shared generation per pair). Deterministic scoring produced contingency counts n11 = 120, n10 = 0, n01 = 0, n00 = 0 (baseline = 120/120; guarded = 120/120). Paired difference (A - B) = 0.00; paired-bootstrap 95% CI = [0.00, 0.00] (10,000 resamples, seed 1729); exact McNemar two-sided p = 1.0. Because generation was intentionally shared, these outcomes measure validator-acceptance parity for identical candidates rather than independent generator or repair robustness. We provide artifact-grounded evidence (execution and analysis manifests, per-task validator traces), an explicit evidence-to-claim mapping table, a nearest-work comparison matrix, and preregistered protocol modifications required to obtain a discriminating head-to-head comparison in follow-ups.

I. INTRODUCTION

Automating intent→configuration translation and safe sequencing of multi-vendor NetOps changes is operationally critical: production networks require both correctness guarantees (reachability, isolation, absence of transient safety violations) and flexibility to express diverse intents across vendor CLIs. Two architectural approaches are common: translation-first pipelines that restrict generation to a vendor DSL and apply formal verification/solvers, and agentic generate→validate→repair pipelines that centralize generation in an LLM and rely on deterministic validators plus bounded repair. Prior literature emphasizes both deterministic guarantees (formal verification) and the promise of LLM-driven flexibility, but direct, preregistered comparisons that produce artifactized per-task validator traces are rare.

This study’s preregistered head-to-head question was: for intent-driven network changes across heterogeneous vendor

CLIs and realistic emulator topologies, which architecture yields higher end-to-end operational correctness (measured by deterministic validation: reachability and policy isolation) — (A) translation-first or (B) agentic generate→validate→repair? The preregistration (CAND-2026-001-prereg-v1) fixed a paired design (N = 120 paired tasks), the primary estimand (paired difference in binary end-to-end correctness A - B), the generator (gpt-5-mini), the deterministic validator (deterministic_netops_validator_v5), and the analysis executor (paired_binary_analysis_v1).

A decisive preregistered design choice was shared_initial_candidate semantics: each paired task used a single LLM candidate generated once and supplied identically to both architectures. This choice deliberately removes generator variance from the confirmatory estimand, turning the experiment into a validator-acceptance parity test for identical candidates rather than an independent generator or repair efficacy comparison. The remainder of the paper (Methods, Results, Discussion) makes that restriction explicit and maps preregistered hypotheses to the measured estimand with artifact references.

II. RELATED WORK

Two research traditions frame our contribution and inform the experiment design. First, formal translation→verification systems focus on deterministic correctness and rely on programmatic abstractions and solver support to produce or certify device configurations. Foundational work on programmatic network verification—SymNet and related abstractions—models network device behavior and forwarding semantics to enable automated reachability analysis and to scale verification via program transformations and symbolic evaluation [10,11]. Complementary work such as "Don’t Mind the Gap" highlights the semantic and implementation gaps that can arise between high-level specification and low-level device behavior and motivates careful abstraction and incremental verification to preserve correctness across translations [10]. More recent formal languages and algebraic frameworks (e.g., Weighted NetKAT and related quantitative verification

approaches) supply compositional semantics and solver interfaces that enable constraint-guided synthesis and metric-aware verification; these tools underpin translation-first pipelines that convert constrained DSL outputs into verifier-checked device artifacts and, when combined with solvers, can produce provably safe configurations under the modeled semantics [5]. Collectively, this literature shows that translation-first architectures trade broader expressivity for stronger, analyzer-level guarantees: by restricting output space and using formal semantics, they can reduce downstream ambiguity and make acceptance decisions deterministic and auditable.

Second, LLM-driven and agentic AIOps work emphasizes flexible, human-friendly intent expression and iterative correction. Recent intent-translation and agentic NetOps studies (for example, INTA at ICNP 2025) investigate conversational intent parsing, in-context translation strategies, and the role of sampling/selection when LLMs produce multiple candidate configurations; these studies document generator variance and the need for downstream validation/repair instrumentation to bound unsafe proposals [3]. Parallel work on verifier-assisted agentic pipelines (e.g., Mahmood & Song, IFIP Networking 2026) demonstrates multi-agent orchestration patterns in which LLM components produce candidates that are iteratively checked by deterministic validators and revised until acceptance; such self-correcting pipelines make the validator the operational authority, instrumenting repair loops and logging verifier traces for auditability [4]. Empirical work on LLM hallucination reduction and structured grounding further argues for hybrid generate→validate→repair designs where validators and constrained prompts limit output-space errors while repair loops address recoverable discrepancies [3,4].

Comparative axes and residual gaps. Prior works differ along a compact set of experimental axes relevant to operational trust: (1) independent generation sampling (whether experiments exercise LLM sampling variance by producing independent candidates per arm), (2) repair-loop instrumentation (whether repair attempts are instrumented and counted), (3) validator acceptance focus (whether evaluation privileges deterministic, formal acceptance criteria versus softer human-centric measures), and (4) emulator/hardware realism (whether experiments validate behavior on real devices or emulators). Formal translation→verification systems excel on axis (3) and (4) when combined with high-fidelity device models, but they commonly restrict generation expressivity and therefore do not explore generator variance on axis (1). Agentic LLM pipelines naturally explore axis (1) and (2) but depend heavily on validator design to bound unsafe outputs. Importantly, few prior studies provide preregistered, paired experimental designs with archived per-task deterministic validator traces that permit reproducible, auditable tests of downstream acceptance parity given identical candidates.

Positioning of the present run. The experiment reported here intentionally targets the intersection of these traditions: it uses LLM generation but freezes generator variance via `shared_initial_candidate` semantics so that the confirmatory estimand isolates downstream validator-acceptance behavior.

This choice draws on the formal literature’s emphasis on deterministic acceptance criteria (we operationalize them via `deterministic_netops_validator_v5`) while adopting artifactized provenance and per-task traces common to reproducible agentic AIOps evaluations. By archiving per-task validator traces, contingency tables, and reconciliation artifacts, the run complements prior translation-first and agentic studies: it does not resolve generator robustness or repair coverage questions under independent sampling (those remain unobserved here), but it supplies a reproducible validator-parity measurement and a concrete protocol that can be modified (e.g., remove `shared_candidate`, enable repair instrumentation, increase `model_calls_used`) to exercise the axes left untested by prior work.

III. METHODOLOGY

Preregistration and execution contract. The preregistration (`CAND-2026-001-prereg-v1`) fixed the primary estimand (paired difference in binary end-to-end correctness $A - B$), `N_confirmatory` = 120 paired tasks, `generator` = `gpt-5-mini`, `validator` = `deterministic_netops_validator_v5` (v5.0), `adapter_family` = `hosted_netops_gvr_v1`, `maximum_model_calls` = 240, and `shared_initial_candidate` semantics. The preregistration also specified `paired_binary_analysis_v1` (bootstrap percentile CIs and exact McNemar test) and a power plan targeting a ~0.15 paired difference with $N=120$. The preregistration artifact is archived and its checksum is recorded in the execution manifest.

Task generation and difficulty calibration. Confirmatory tasks (120 pairs) were generated by `deterministic_netops_task_generator_v5`. Each manifest entry (`execution/task_manifest.jsonl`) contains: intent text, canonical reference answer, initial and expected final states, required safe sequences, and difficulty metadata (`changed_interface_count`, `dependency_rule_count`, level: medium/hard/very_hard). Representative entries (e.g., task-000001) and their canonical reference answers are archived and used by the deterministic validator as ground truth for normalized comparison.

Generation and shared-candidate semantics. The orchestration implemented `shared_initial_candidate`: for each pair the system made a single LLM call and distributed the identical textual candidate to both downstream arms. Execution accounting (`execution/execution_manifest.json`) records `planned_episode_count` = 240, `completed_episode_count` = 240, and `model_calls_used` = 120 (one shared generation per pair). Provider audit (`analysis/deterministic_reconciliation.json`) reports `hosted_model_call_event_count` = 120 and `cross_condition_cache_key_reuse_observed` = false, consistent with execution isolation.

Validator rules and scoring. The `deterministic_netops_validator_v5` applies `full_intent_constraint_validation`: syntactic normalization, required command inclusion, ordering/dependency checks (make-before-break, shutdown→modify→restore),

final state consistency, and emits `validation_after.valid` (boolean). A configuration is scored success (1) only when `validation_after.valid == true`. Per-task scoring JSONs (`execution/scoring/task-000XXX-*.json`) record `normalized_response`, `parsed_command_count`, `required_command_count`, `validation_before/after` snapshots, `repair_applied` flags and the scorer id/version. These files are the canonical evidence for every per-task decision and are archived in the execution bundle.

Analysis plan and exact procedures. Analysis followed `paired_binary_analysis_v1`, computing contingency counts (`n11, n10, n01, n00`), paired difference $A - B$, bootstrap percentile 95% CI (10,000 resamples, `bootstrap_seed = 1729`), and exact McNemar two-sided test. Analysis artifacts include `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl` (bootstrap parameters), and `analysis/deterministic_reconciliation.json`. Secondary estimands (repair coverage, mean repair iterations) were preregistered; their measurement depends on nonzero `repair_applied` flags in scoring JSONs.

IV. RESULTS

Execution accounting and primary numeric outcomes. The execution manifest records `planned_episode_count = 240`, `completed_episode_count = 240`, `failed_episode_count = 0`, `model_calls_used = 120` (one shared LLM call per paired task), and `maximum_model_calls = 240`. Deterministic reconciliation (`analysis/deterministic_reconciliation.json`) confirms `complete_pair_count = 120` and provider audit counts: `hosted_model_call_event_count = 120`, `completed_hosted_model_call_event_count = 120`, `failed_hosted_model_call_event_count = 0`, `cache_miss_count = 120`, `cache_hit_count = 0`, and `cross_condition_cache_key_reuse_observed = false`. These provider-audit numbers substantiate that the orchestration created exactly one independent generation per pair and maintained execution isolation as preregistered.

Primary contingency and statistical summaries. The analysis contingency (`analysis/paired_contingency_table.csv`) is `n11 = 120`, `n10 = 0`, `n01 = 0`, `n00 = 0`. Marginal success rates are `baseline = 120/120 = 1.0` and `guarded = 120/120 = 1.0`; the observed paired difference $A - B = 0.00$. The paired bootstrap percentile 95% CI was computed with `bootstrap_resamples = 10,000` and `bootstrap_seed = 1729` (`analysis/analysis_log.jsonl`) and returned `[0.00, 0.00]`. The exact McNemar two-sided test yields $p = 1.0$ (test applied to the observed discordant counts `n10` and `n01`). These statistics are deterministic given the degenerate contingency table.

Repair and secondary outcomes. Across all 240 episodes the archived per-task scoring traces record `repair_applied = false` (see `execution/scoring/*.json`). Aggregation yields `repair_applied count = 0`; consequently preregistered secondary estimands that quantify repair coverage and mean repair iterations are unobserved in this execution. The absence of repair activity is present in every representative scoring JSON (for

example, `execution/scoring/task-000001-baseline.json` records `repair_applied: false` and `validator_trace.validation_after.valid: true`), and is consistent across the artifact set.

Representative per-task diagnostics. Representative scoring artifacts show that `shared_initial_candidate` matched the `normalized_reference_answer` modulo command ordering in several cases and that validators parsed and validated required commands: example excerpts from archived scoring files illustrate typical complexity and validator checks—

- `task-000001`: `parsed_command_count = 4`, `required_command_count = 4`, `validation_after.valid = true`; `normalized_configuration` and `final_state` fields show the expected interface `admin/mtu/vlan` values and no violations (`execution/scoring/task-000001-*.json`).
- `task-000002`: `parsed_command_count = 2`, `required_command_count = 2`, `validation_after.valid = true`; difficulty flagged as `level = hard` with `pattern = make_before_break_uplink_switchover`.
- `task-000003`: `parsed_command_count = 6`, `required_command_count = 6`, `validation_after.valid = true`; difficulty = `hard` with `pattern = paired_link_atomic_mtu_change`.

These representative entries demonstrate that the validator applied dependency and ordering checks (`make-before-break`, `shutdown→modify→restore`) and that the normalized responses satisfied those constraints for the shared candidates.

Contamination and missingness diagnostics. The preregistered contamination checks produced no flags: `analysis/contamination_summary.csv` is empty. Missingness diagnostics (`analysis/missingness_summary.csv`) report `complete_pairs = 120` and zero for `baseline_only_observed_pairs`, `guarded_only_observed_pairs`, `both_missing_pairs`, and `failed_episodes`. Provider audit and execution logs show no retries or infrastructure failures beyond the planned single generation per pair; `execution/execution_manifest.json` and `analysis/deterministic_reconciliation.json` contain the machine-readable accounting.

Interpretation constrained to measured estimand. Because the design used `shared_initial_candidate` semantics and a single deterministic validator per arm, the numerical result—perfect concordance across all 120 paired tasks—directly measures validator-acceptance parity for identical LLM candidates. The degenerate contingency produces a point mass estimate (difference = 0.00) and a degenerate bootstrap CI and p-value; these are accurate descriptions of the observed data. However, by design these results do not address independent generator sampling variance, multi-sample selection, or the effectiveness of repair loops when the generator is called separately for each arm. Those secondary dimensions remained unexercised here and are therefore unobserved in the artifacts.

V. DISCUSSION

What the run measures — and what it does not. Because `shared_initial_candidate` semantics were used, the preregistered confirmatory estimand reduces to validator-acceptance parity: given identical LLM candidates, do downstream orchestration and verification paths differ in deterministic val-

validator acceptance? The observed complete concordance ($n_{11} = 120$) answers this restricted question: both validators accepted the identical candidates for every paired task. The run does not exercise independent generator variance, sampling effects, or repair efficacy; therefore claims about which architecture would produce more correct initial candidates or achieve higher repair coverage when generating independently are untested.

Evidence-to-claim mapping (compact table). Table 2 (in manuscript) maps preregistered hypotheses/estimands to measured observables and archived artifacts. Example mappings: H1 (paired difference $A - B$) $\rightarrow$ measured estimand = validator-acceptance parity under shared_initial_candidate; supporting artifacts: analysis/paired_contingency_table.csv and analysis/condition_summary.csv ($n_{11}=120, n_{10}=0, n_{01}=0, n_{00}=0$); verdict: tested but restricted. H2 (agentic repair coverage advantage) $\rightarrow$ observed repair_applied count = 0 in execution/scoring/*.json; verdict: unobserved in this execution.

Operational implications and protocol guidance for discriminating comparisons. The artifact bundle documents a reproducible protocol for validator regression and orchestrator equivalence checks under fixed candidates. To test independent-generation and repair differences the preregistered protocol must be modified (and these modifications can be preregistered and audited): (i) remove shared_initial_candidate so each arm receives an independent LLM generation per pair (model_calls_used would rise from 120 to 240 for initial candidates); (ii) allow multiple sampled initial generations per arm to estimate generator variance; (iii) enable and instrument repair iterations so repair_applied counters become informative; (iv) increase emulator fidelity or include hardware runs to reduce simulator-to-hardware contamination risk. Each modification yields verifiable changes in execution artifacts (model_calls_used, hosted_model_call_event_count, nonzero repair_applied flags) and therefore supports reproducible discrimination across architectures.

Threats to validity (concise). Internal validity: shared_candidate removes generator variance by design; this is a feature not a flaw but narrows the estimand. Construct validity: deterministic validator acceptance operationalizes a conservative correctness definition but omits human-facing properties (explainability, maintainability). External validity: synthetic tasks and Mininet-derived emulation plus a single model (gpt-5-mini) and single validator (deterministic_netops_validator_v5) limit generality. Conclusion validity: degenerate concordance (all pairs concordant) produces point mass CI and $p = 1.0$; these statistics correctly describe the observed data but do not imply production-level equivalence under independent operation.

Reproducibility and audit artifacts. All primary artifacts are archived and referenced by execution/execution_manifest.json: preregistration (CAND-2026-001-prereg-v1), execution/model_configuration.json (declared_model_version

= gpt-5-mini), execution/task_manifest.jsonl, execution/responses/*, execution/scoring/*.json, analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/analysis_log.jsonl, and analysis/deterministic_reconciliation.json. Per-task validator traces (execution/scoring/task-000XXX-*.json) are the canonical records for each scored decision. The analysis executor parameters (bootstrap_resamples = 10000, bootstrap_seed = 1729) are recorded in analysis/analysis_log.jsonl and enable deterministic recomputation of the reported CI.

VI. LIMITATIONS

- Preregistered shared_initial_candidate semantics were used; this intentionally removes generator variance and prevents this run from assessing independent generation robustness differences between architectures.
- The task corpus was synthetic and executed in an emulator (Mininet-derived topologies and public vendor example grammars); emulator-to-hardware divergence limits external validity to production devices.
- A single LLM (gpt-5-mini) and a single deterministic validator (deterministic_netops_validator_v5) were used; results do not imply multi-model or multi-validator generality.
- No repairs were applied in this run (repair_applied = false in per-task traces); preregistered secondary estimands about repair coverage remain unobserved for this execution.
- The binary deterministic validator acceptance metric does not capture operator-facing properties such as explanation quality, maintainability, or human trust, which matter in production but were outside the metric used here.

VII. CONCLUSION

We executed a preregistered, artifactized paired experiment comparing translation-first and agentic generate $\rightarrow$ validate $\rightarrow$ repair NetOps pipelines under shared_initial_candidate semantics. For 120 paired tasks deterministic validators accepted identical LLM candidates in both arms for every pair ($n_{11} = 120, n_{10} = n_{01} = n_{00} = 0$), yielding paired difference 0.00 (bootstrap 95% CI [0.00, 0.00], McNemar $p = 1.0$). No repairs were applied (repair_applied = false for all episodes); preregistered hypotheses about repair coverage are therefore unobserved. The run precisely measures validator-acceptance parity under controlled shared candidates and provides a reproducible artifact bundle and explicit protocol modifications for future discriminating comparisons that exercise independent generation and repair coverage.

DISCLOSURE STATEMENT

This research was executed autonomously after the autonomy lock under the frozen master prompt archived with the run provenance. Preregistration: CAND-2026-001-prereg-v1 (archived and used to freeze

the design; preregistration_sha256 recorded in the execution manifest). Generation used gpt-5-mini (declared_model_version in execution/model_configuration.json); provider record 'openai_responses'. The hosted adapter family was hosted_netops_gvr_v1. Deterministic artifacts included deterministic_netops_task_generator_v5 and deterministic_netops_validator_v5 (validator_version 5.0). The run deliberately used the preregistered shared_initial_candidate generation semantics: both pipeline arms received the same initial LLM candidate per task; execution accounting shows model_calls_used = 120 (one shared generation per pair) while planned episodes were 240. Primary analysis used paired bootstrap percentile CIs (10,000 resamples, seed 1729) and an exact McNemar two-sided test executed by paired_binary_analysis_v1. No human manually edited, rewrote, or post-processed technical content after the autonomy lock. The Research Director selected the broad topic family and constrained the initial master prompt prior to the autonomy lock; the autonomous run performed literature review, final research-question selection, preregistration, code generation, experiment execution, analysis, peer review, and manuscript drafting after the lock. Immutable reference to the initial master prompt: provenance/master_prompt.txt (SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). The full execution and analysis artifact bundle is archived and referenced by the execution manifest included with this submission.

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